

Robotic Control in Space and In-Space Assembly

A ten-paper digest with claim locations and full bibliography | distilled from *notes/literature-review.md* (501 arXiv records, 86 core space-domain pose-estimation papers) | compiled 2026-09-24. Every section, table and figure reference below was checked against the arXiv full text of the version cited in the bibliography.

How to read this document

Each paper gets one card. The left column is the claim you would cite; the right column tells you exactly which section, table or figure carries it, so you can verify in under a minute rather than re-reading the paper. Papers 1–4 are the **control and assembly** spine; papers 5–8 are the **perception-under-compute-budget** spine; papers 9–10 are the two **reality checks** (know when you are wrong; beat the cheap baseline). Reading order if time is short: 1, 5, 3, 9.

The state of the field in seven sentences

Dimension	Where the field is in 2026
Task framing	Nearly all published work targets a non-cooperative, tumbling target in rendezvous or servicing. Cooperative, self-designed docking ports in a repeated assembly cycle are almost unstudied (1 core paper: #4).
Controller of record	Classical position- and image-based visual servoing still dominate, with MPC (often robust/tube MPC) as the tracking layer of choice (#1, #2).
Learned control	Now flight-proven but only inside a monitored envelope with a classical fallback (#3).
Perception	Monocular CNN/ViT keypoint → PnP is the standard pipeline; synthetic-only training now reaches state of the art on hardware-in-the-loop imagery (#5).
Compute budget	Settled question: 178× smaller costs $\leq 4\times$ accuracy, and modular quantization can make a 4×-smaller model <i>more</i> accurate (#6, #7). Do not pose 'size vs. accuracy' as a research question.
Open gap	Uncertainty. Only ~5 of 86 core space papers quantify it, yet a cheap ensemble gate took a real grasp success rate from 35% to 90% (#8, #9).
Honest baseline	A well-tuned fiducial marker on a flat, self-designed port is near-perfect. Curvature, not the algorithm, is what breaks markers (#10).

Scope warning

The underlying search was **arXiv-only**: no IEEE Xplore, AIAA ARC, *Acta Astronautica*, *JGCD*, or i-SAIRAS/ASTRA proceedings, which is where much space-robotics control work appears. No citation-graph ranking was possible, so 'top 10' here means *most useful for orienting yourself*, weighted toward flight-validated control and assembly results, not most-cited. Numbers quoted are the authors' own, read from the cited section or table.

Part I — Control and assembly: what actually works in orbit

1. Amaya-Mejia et al. 2024 - Visual Servoing for Robotic On-Orbit Servicing: A Survey

IEEE iSpaRo 2024 | arXiv:2409.02324 | DOI 10.1109/iSpaRo60631.2024.10687516

Why it matters: Read first: vocabulary and phase decomposition for any space control section.

Key claim / finding	Where to find it in the paper
Organises all on-orbit servicing visual servoing work into three mission phases — Recognition, Approach, Contact . This decomposition is the field's default framing; use it.	§I Introduction, para. 3 (statement of taxonomy); the review itself is §IV-A / IV-B / IV-C
The space-specific obstacles are enumerated: extreme lighting variation, target specularly, ineffectiveness during eclipse, limited onboard processing, and microgravity-induced tumbling that is hard to reproduce on the ground.	§II Challenges, para. 1–2
Current trends: classical PBVS/IBVS still dominate (PBVS preferred when a geometric model exists), eye-in-hand is the usual camera placement, almost all work targets <i>non-cooperative</i> targets, most testbeds use <i>free-flying</i> rather than <i>free-floating</i> models, and linear controllers remain standard because they map image error to command directly.	§V Discussion (trend paragraphs); per-paper evidence in comparison Tables I (recognition) and II (approach/contact)

2. Doerr, Albee, Ekal, Ventura & Linares 2023 - The ReSWARM Microgravity Flight Experiments

arXiv:2301.01319 (ISS flight results, NASA Astrobee)

Why it matters: The best available evidence on what assembly-grade planning and control actually does in orbit.

Key claim / finding	Where to find it in the paper
Three capabilities demonstrated on the ISS for the first time : (1) dynamics-aware sampling-based global planning, (2) information-aware planning with online inertial model learning, (3) tube-based robust tracking under model uncertainty.	§1 Introduction (contribution bullets); restated in §7 Conclusion; results per capability in §6.1, §6.2, §6.3
Defines the canonical assembly duty cycle: a free-flyer grapples a payload, which changes system mass and inertia , which then corrupts both planning and control — so the model must be re-estimated in the loop.	§3.3 The On-Orbit Assembly Scenario (scenario); dynamics and uncertainty formalism in §3.1–§3.2
Robust tube MPC is the tracking layer; RATTLE is the combined planner/estimator/controller.	§4.5 Robust Model Predictive Control; §4.6 RATTLE; integration in §4.7
Compute is the binding constraint . Robust tube MPC replanned in 0.160 ± 0.050 s against a 0.2 s budget, and on-orbit localisation load made everything roughly 2× slower than ground tests .	§6.4 Planning and Control Speeds on Resource-Constrained Hardware, with Fig. 28 and Table 4

3. Chapin, Stewart, Leontie & Henshaw 2025 - APIARY: Autonomous Planning In-space Assembly RL Free-Flyer

IEEE iSpaRo 2025 (Best Paper, Orbital Robotics) | arXiv:2512.03729 | DOI 10.1109/iSpaRo66239.2025.11436341

Why it matters: The single most important recent data point on learned control in space.

Key claim / finding	Where to find it in the paper
First reinforcement-learning control of a free-flyer in space (NASA Astrobee, ISS, 27 May 2025). A 6-DOF PPO actor-critic policy trained in NVIDIA Isaac Lab, randomised over goal pose and mass.	Abstract and §I Introduction (claim of first); training setup in §II-A Training: NVIDIA Omniverse
Sim → ground → flight pipeline (Isaac Lab → Astrobee Gazebo simulator → Granite Lab air-bearing table → ISS) is presented as the reusable contribution, not just the policy.	§II (pipeline overview, Fig. 1); stage results in §III-A (simulation) and §III-B (Granite Lab)
Flight result: 8 maneuvers in ~7 minutes; docking attempt #1 failed, attempt #2 succeeded during loss of signal. On failure, onboard error detection saw position error exceed the policy's threshold and handed control back to the baseline controller , holding pose safely.	§IV Flight Testing (maneuver list 1–8 and the failure/recovery narrative), Fig. 7
Takeaway for design: the flight-worthy pattern is <i>learned policy inside a monitored envelope with a classical fallback</i> — supervision, not replacement.	§IV Flight Testing; §V Conclusion

4. Sonawani, Alimo, Detry, Jeong, Hess & Ben Amor 2020 - Assistive Relative Pose Estimation for On-orbit Assembly using CNNs

AIAA SciTech 2020 | arXiv:2001.10673 | DOI 10.2514/6.2020-2096

Why it matters: Still the only paper that squarely couples learned relative pose with on-orbit assembly.

Key claim / finding	Where to find it in the paper
Positions the CNN as assistive to , not a replacement for, a feature-based tracker: it supplies an initial/recovery estimate where hand-crafted features fail.	Abstract; motivation in §1 Introduction, para. 4; restated in §5 Conclusion and Future Work
Architecture sweep on a simulated assembly testbed (80/20 split): VGG-16 13.56° mean rotation / 0.18 m translation → VGG-19 12.21° / 0.15 m → branched VGG-19 8.34° / 0.06 m → parallel VGG-19 4.61° / 0.03 m .	Table 1 (accuracy of all four models); protocol and split in §4 Results, para. 1; dataset in §2 Dataset
Loss design matters: translation and rotation losses are combined with a weight β tuned so both terms are comparable at convergence ($\beta = 10$ here).	§3.1 Loss Function
Caveat before citing: 2020-era regression CNN, synthetic-only, no closed-loop control and no uncertainty output. Treat it as the framing precedent, not the method to copy.	§4 Results (error heatmaps, Figs. 7–8); §5 Conclusion and Future Work

Part II — Perception inside a spacecraft compute budget

5. Park & D'Amico 2024 - Bridging the Domain Gap for Flight-Ready Spaceborne Vision (SPNv3)

J. Spacecraft and Rockets | arXiv:2409.11661 | DOI 10.2514/1.A36130

Why it matters: The methodology paper for synthetic-only training that survives real imagery.

Key claim / finding	Where to find it in the paper
Writes down explicit requirements for a spaceborne network : run within the ADCS update rate (0.5 Hz cited from the CPOD mission), minimise learnable parameters, and stay batch-agnostic so single-image inference is valid.	§3.4 Summary (the three requirements); CPOD 0.5 Hz / 10 Hz evidence in §3.1; GPU flight heritage in §3.1.1
Trained on synthetic images only , a mid-size SPNv3 variant would have ranked first in the SPEC2021 lightbox category — beating entries that used the unlabeled real test images during training.	§1 Introduction (claim); numbers vs. state of the art in §6.4.1, Table 8 (SPNv3-M, 39.6M params)
≤40 ms per inference on an NVIDIA Jetson Nano 4GB ; the latency table shows ViT backbones beating CNNs at equal accuracy (ViT-S/16, 22.7M params, ~22 ms test latency).	§1 Introduction (40 ms); measured table in §6.3.2 Latency on NVIDIA Jetson Nano, Table 3
What buys robustness, ranked by ablation: heavy data augmentation, better pretraining/transfer learning, higher input resolution, normalisation-layer choice.	Augmentation list Table 1 (§5.2); ablations §6.3.1 (architecture), §6.3.3 (norm layers), §6.3.4 (transfer), §6.3.5 (resolution), §6.3.6 (augmentation)
Honest limitation: results depend on an ImageNet-pretrained ViT backbone, and <i>how</i> it was pretrained changes downstream robustness — a barrier if you cannot afford that pretraining.	§6.5 Limitations

6. Posso, Bois & Savaria 2022 - Mobile-URSONet: an Embeddable Neural Network for Onboard Spacecraft Pose Estimation

IEEE ISCAS 2022 | arXiv:2205.02065 | DOI 10.1109/ISCAS48785.2022.9937721

Why it matters: The paper that already answered 'how much accuracy does shrinking cost?'

Key claim / finding	Where to find it in the paper
178× fewer parameters while degrading the ESA pose score by no more than 4× , i.e. 139× better accuracy-per-parameter than URSONet.	Abstract; §I Introduction, para. 3; measured in §III-B Results (the 8-bin model)
Where the cost lands is not uniform: position error tracks only backbone size (MobileNetV2 has 13× fewer parameters than ResNet-101, position error degrades ≤3×), while orientation error tracks the size of the orientation head.	§II-B Neural network architecture (parameter counts); §III-B Results (error attribution)
Soft classification of orientation beats direct regression, but more bins per axis overfits the synthetic domain — an early warning that capacity and sim2real interact.	§II-C Loss functions; §III-B Results
Introduces an explicit generalization metric (synthetic vs. real score ratio) so compression and domain gap are not conflated.	§II-A Generalization metric

7. Javed, Li, Price, Hu & Salzmann 2024 - Modular Quantization-Aware Training for 6D Object Pose Estimation (MQAT)

Transactions on Machine Learning Research (TMLR) 2024 | arXiv:2303.06753

Why it matters: Kills the assumption that smaller is always worse.

Key claim / finding	Where to find it in the paper
>7% accuracy gain (ADI-0.1d) over the full-precision baseline while shrinking the model $\geq 4\times$. Quantization acted as a regulariser rather than a tax.	Abstract; measured comparison in §4.3.3 MQAT Compared to Uniform and Mixed QAT (tables); restated §6 Conclusion
Mechanism: exploit the modular structure of pose networks (backbone / FPN / heads) and assign per-module bit widths in a graduated sequence, instead of one uniform precision.	§3.1 Modular Architecture for 6D Pose Estimation; §3.2 MQAT Algorithm
Module sensitivity is measurable and uneven — the feature-pyramid fusion is the fragile part.	§4.3.2 Quantized FPN Sensitivity Study; §4.3.1 MQAT Quantization Flow
Generalises across datasets, architectures and QAT algorithms, including the SPEED+ spacecraft benchmark.	§4.4.1 Dataset and QAT Generality; §4.4.2 Architecture Generality; SPEED+ results in Appendix A.3
Read the caveat: compression ratio is not a measured hardware speed-up; latency was not benchmarked on an accelerator.	§5 Limitations and Discussion, 'Latency.' paragraph

8. Ali Ousalah, Kacem, Ghorbel, Koumandakis & Aouada 2025 - Uncertainty-Aware Knowledge Distillation for Compact 6DoF Pose Estimation

IEEE/RSJ IROS 2025, pp. 15398-15405 | arXiv:2503.13053 | DOI 10.1109/IROS60139.2025.11245982

Why it matters: Unifies 'small model' and 'knows when it is wrong' in one training objective.

Key claim / finding	Where to find it in the paper
Distillation is weighted by the teacher's per-keypoint uncertainty , so the student copies confident keypoints and discounts unreliable ones.	§IV-A Uncertainty-Aware Prediction-Level KD; formal setup in §III Problem Statement
The same uncertainty also selects <i>where</i> in the feature maps knowledge is transferred.	§IV-B Prediction-related Feature-per-Keypoint KD
Deliberately uses epistemic (model) uncertainty only, so distillation is not driven by data noise.	§IV-A (justification paragraph)
On SPEED+ with a small SPNv2 ($\phi=0$) student: baseline student lightbox pose error 0.368 $\rightarrow$ 0.257 , against a 0.202 teacher — i.e. most of the teacher's real-domain robustness recovered in a far smaller model (teacher EfficientDet-D6 57.8M vs. student D0 3.8M).	Table IV (§V-C Spacecraft Pose Estimation with SPEED+); model sizes in Table I (§V-A); LINEMOD results Tables II–III (§V-B)

Part III — Two reality checks

9. Shi, Zhu, Tremblay, Birchfield, Ramos, Anandkumar & Zhu 2021 - Fast Uncertainty Quantification for Deep Object Pose Estimation

IEEE ICRA 2021 | arXiv:2011.07748

Why it matters: The cheapest usable confidence gate; the closed-loop payoff is huge.

Key claim / finding	Where to find it in the paper
Method in one line: run 2–3 pre-trained pose estimators (different architectures or training data) and use their average pairwise disagreement as the uncertainty signal. No retraining, one forward pass per model.	§I Introduction (contributions); §III-B Heterogeneous Ensemble for Uncertainty Quantification
Of four disagreement metrics, ADD is the best learning-free one and nearly matches a learned metric that needs labeled target data (Spearman correlation with true error: ADD ensemble 0.64–0.68 vs. 0.26–0.40 for network confidence).	§III-C Learned Disagreement Metric; correlations in Tables I and II (§V-A Correlation Analysis)
Used as a viewpoint gate , it picks the frame to trust: 3.5 cm mean ADD error vs. 4.9 cm for raw network confidence.	§V-B Application I: Camera Perspective Selection, Table III
Closed-loop payoff: real-robot stable grasp rate 35% $\rightarrow$ 90% , failures 55% $\rightarrow$ 0%. Same argument transfers to gating a docking approach.	§V-C Application II: Real-World Robotic Grasping, Table IV
Explicitly motivated by sim2real overconfidence — estimators are unreliable exactly where the input leaves the training domain.	§I Introduction, para. 1–2

10. Thakur, Vrba & Saska 2026 - AstraTag: Spacecraft Fiducial Marker for Autonomous Rendezvous, Proximity Operations, and Docking

arXiv:2606.27566 (preprint)

Why it matters: The baseline any learned docking estimator must beat. Answer it explicitly.

Key claim / finding	Where to find it in the paper
Motivating failure mode is precisely the docking one: existing single-scale markers leave the camera field of view at close range , exactly when tracking matters most.	Abstract; §I Introduction, para. 2; prior-marker analysis in §II Related Work
Design: recursive, self-similar square Spidron template detectable across scales, identified by a 48-bit signature with Generalised Reed-Solomon coding.	§III-A AstraTag (template); §III-B Encoding; pipeline §III-C Detection; §III-D Pose Estimation
For markers on curved spacecraft skins, a thin-plate-spline re-warp fallback uses the marker's internal rectangular borders as extra correspondences, extending detection to 70° out-of-plane.	§I Introduction (contribution bullets); mechanism in §III-C Detection
Curved-surface benchmark: AstraTag holds 97–100% detection from 10° to 70° ; three-layer Fractal ArUco manages 93–98% at moderate angles but collapses to 2% at 40° and averages 21.9%, usable only within 50 cm.	§IV-C1 Space Station Dataset, Fig. 9; marker print geometry in Table I (§IV-B)
On a flat surface all three markers are near-perfect (Fractal ArUco recovers to 99.3% over distance), so surface curvature is the only real differentiator — a cheap flat marker on a port you design is a genuinely strong baseline.	§IV-C2 Spacecraft Dataset, Fig. 10; §IV-D Discussion; §V Conclusions

Bibliography — the ten papers

- [1] Amaya-Mejía, L. M., Ghita, M., Dentler, J., Olivares-Mendez, M., and Martinez, C. “Visual Servoing for Robotic On-Orbit Servicing: A Survey.” *2024 International Conference on Space Robotics (iSpaRo)*, IEEE, 2024. doi:10.1109/iSpaRo60631.2024.10687516. arXiv:2409.02324.
- [2] Doerr, B., Albee, K., Ekal, M., Ventura, R., and Linares, R. “The ReSWARM Microgravity Flight Experiments: Planning, Control, and Model Estimation for On-Orbit Close Proximity Operations.” arXiv:2301.01319, 2023 (rev. 16 Jul 2023).
- [3] Chapin, S., Stewart, K., Leontie, R., and Henshaw, C. G. “Autonomous Planning In-space Assembly Reinforcement-learning free-flyer (APIARY) International Space Station Astrobee Testing.” *2025 International Conference on Space Robotics (iSpaRo)*, IEEE, 2025 (Best Paper Award, Orbital Robotics). doi:10.1109/iSpaRo66239.2025.11436341. arXiv:2512.03729.
- [4] Sonawani, S., Alimo, R., Detry, R., Jeong, D., Hess, A., and Ben Amor, H. “Assistive Relative Pose Estimation for On-orbit Assembly using Convolutional Neural Networks.” *AIAA SciTech 2020 Forum*, 2020. doi:10.2514/6.2020-2096. arXiv:2001.10673.
- [5] Park, T. H., and D’Amico, S. “Bridging the Domain Gap for Flight-Ready Spaceborne Vision.” *Journal of Spacecraft and Rockets*, 2025. doi:10.2514/1.A36130. arXiv:2409.11661 (v2, 3 Sep 2025).
- [6] Posso, J., Bois, G., and Savaria, Y. “Mobile-URSONet: an Embeddable Neural Network for Onboard Spacecraft Pose Estimation.” *2022 IEEE International Symposium on Circuits and Systems (ISCAS)*, 2022. doi:10.1109/ISCAS48785.2022.9937721. arXiv:2205.02065.
- [7] Javed, S., Li, C., Price, A., Hu, Y., and Salzmann, M. “Modular Quantization-Aware Training for 6D Object Pose Estimation.” *Transactions on Machine Learning Research*, 2024. arXiv:2303.06753 (v3, 4 Nov 2024).
- [8] Ali Ousalah, N., Kacem, A., Ghorbel, E., Koumandakis, E., and Aouada, D. “Uncertainty-Aware Knowledge Distillation for Compact and Efficient 6DoF Pose Estimation.” *Proc. 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 15398–15405, 2025. doi:10.1109/IROS60139.2025.11245982. arXiv:2503.13053.
- [9] Shi, G., Zhu, Y., Tremblay, J., Birchfield, S., Ramos, F., Anandkumar, A., and Zhu, Y. “Fast Uncertainty Quantification for Deep Object Pose Estimation.” *2021 IEEE International Conference on Robotics and Automation (ICRA)*, 2021. arXiv:2011.07748.
- [10] Thakur, R. K., Vrba, M., and Saska, M. “Spacecraft Fiducial Marker for Autonomous Rendezvous, Proximity Operations, and Docking.” arXiv:2606.27566, 2026 (preprint).

Bibliography — supporting works worth one paragraph each

- [11] Sharma, S., Beierle, C., and D’Amico, S. “Pose Estimation for Non-Cooperative Spacecraft Rendezvous Using Convolutional Neural Networks.” *2018 IEEE Aerospace Conference*, 2018. arXiv:1809.07238. — **Origin of the SPEED dataset lineage**; cite for context. Key claims: training-set size correlates with accuracy, noise-augmented training transfers to noisy test images, transfer learning suffices (all four findings enumerated in §IV Conclusions).
- [12] Posso, J., Bois, G., and Savaria, Y. “Real-Time Spacecraft Pose Estimation Using Mixed-Precision Quantized Neural Network on COTS Reconfigurable MPSoC.” *IEEE NEWCAS 2024*. arXiv:2407.06170. — Mobile-URSONet on a Xilinx ZCU104 FPGA: **58.7 FPS at 0.865 W (67.9 FPS/W)**, 7.7× faster and 19.5× more energy-efficient than prior reported values (Tables II–III, §III-D / §IV-A). Open source. Note the cost: ESA score 0.296 → 0.411 after mixed-precision quantization (Table I).
- [13] Doerr, B., Albee, K., Ekal, M., Linares, R., and Ventura, R. “Safe and Uncertainty-Aware Robotic Motion Planning Techniques for Agile On-Orbit Assembly.” arXiv:2102.10348, 2021. — The LQR-RRT* + robust tube MPC formulation that ReSWARM later flew; read §4.5 (‘Implications for On-Orbit Assembly Robustness’).
- [14] Schwartz, D., Hansson, A., Bodmer, S., Sternberg, D., Jia-Richards, O., and Albee, K. “SmallSatSim: A GPU-Accelerated Microgravity Robotics Toolkit for Planning, Control, and Policy Learning.” arXiv:2603.14598, 2026. — MuJoCo/MJX testbed putting model-based and learned controllers on one problem definition; includes contact-rich rendezvous and docking tasks.
- [15] Laurent, G. J., and Sandoz, P. “FMAC: a Fair Fiducial Marker Accuracy Comparison Software.” arXiv:2601.07723, 2026. — Ready-made, physically-based ray-traced evaluation protocol (distortion, defocus, diffraction blur) for benchmarking a marker baseline honestly. Adopt it rather than inventing a protocol.

What is missing from this list (deliberately)

Contact dynamics and grasping of tumbling targets, event-camera and neuromorphic sensing, low-light image enhancement, free-floating manipulator dynamic coupling, and multi-agent construction-site planning each have active literatures omitted here for length. Survey [1] §IV-C covers contact; the parent review (*notes/literature-review.md*) lists those clusters with identifiers.